

Pocket Ledger

Personal expense tracker dashboard: complete source code

File: index.html · 668 lines · Repository: mohan196/git-test · Generated 28 Sep 2026

Technology stack

Front end	HTML5, hand-written CSS (custom properties, Grid, Flexbox, media queries) and plain JavaScript. No framework.
Charts	Chart.js 4.4.1, loaded from cdnjs.cloudflare.com: stacked bar, donut, line and bar charts with tooltips.
Fonts	Google Fonts: Bricolage Grotesque (headings), IBM Plex Sans (text), IBM Plex Mono (numbers).
Back end	None. The page is fully static and runs entirely in the browser.
Storage	Browser localStorage, key "pocket-ledger-v1": expenses, budgets and currency saved as JSON.

Features

- Summary cards: spent this month (vs. last month), budget remaining with status, daily average with projection, top category
- Six-month stacked bar chart of spending by category
- Donut chart of the month's category split, with amounts and percentages
- Running total vs. budget pace, compared with the previous month
- Per-category budgets with editable limits and On track / Near limit / Over budget labels
- Spending by day of week
- Add-expense form; transactions table with category filter, search and delete
- Month and currency selectors (INR, USD, EUR, GBP); light and dark themes; phone layout

Expense categories

 Housing	 Food & Dining	 Transport	 Utilities
 Shopping	 Health	 Entertainment	 Other

Code layout

Lines	1–11	Document head: fonts and Chart.js
Lines	12–190	CSS: theme, layout and components
Lines	193–316	HTML: page structure
Lines	317–665	JavaScript: data, charts and interactions

To run it, open index.html in any modern browser. An internet connection is needed to load Chart.js and the fonts.

Document head: fonts and Chart.js

```
1 <!doctype html>
2 <html lang="en">
3 <head>
4 <meta charset="utf-8">
5 <meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover">
6 <!-- ARTIFACT-START -->
7 <title>Pocket Ledger</title>
8 <link rel="preconnect" href="https://fonts.googleapis.com">
9 <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
10 <link rel="stylesheet" href="https://fonts.googleapis.com/css2?family=Bricolage+Grotesque:opsz,wght@12..96,
  500;12..96,700&family=IBM+Plex+Sans:wght@400;500;600&family=IBM+Plex+Mono:wght@400;500&display=swap">
11 <script src="https://cdnjs.cloudflare.com/ajax/libs/Chart.js/4.4.1/chart.umd.min.js"></script>
```

CSS: theme, layout and components

```
12 <style>
13 :root {
14   --bg: #f3f5f2;
15   --surface: #fcfcfb;
16   --surface-2: #eefled;
17   --ink: #121715;
18   --ink-2: #4c5652;
19   --muted: #7a8480;
20   --line: #dfe4df;
21   --grid: #e9ece8;
22   --accent: #1d4d45;
23   --accent-ink: #ffffff;
24   --good: #0ca30c; --good-ink: #0a6b0a;
25   --warn: #fab219; --warn-ink: #8a5a00;
26   --bad: #d03b3b; --bad-ink: #b02a2a;
27   --c1: #2a78d6; --c2: #eb6834; --c3: #1baf7a; --c4: #eda100;
28   --c5: #e87ba4; --c6: #008300; --c7: #4a3aa7; --c8: #e34948;
29   --ghost: #b8c0bc;
30   --display: "Bricolage Grotesque", "Segoe UI", system-ui, sans-serif;
31   --sans: "IBM Plex Sans", "Segoe UI", system-ui, sans-serif;
32   --mono: "IBM Plex Mono", ui-monospace, "SFMono-Regular", Menlo, monospace;
33 }
34 @media (prefers-color-scheme: dark) {
35   :root:not([data-theme="light"]) {
36     color-scheme: dark;
37     --bg: #101312; --surface: #1a1d1c; --surface-2: #232826;
38     --ink: #f1f4f2; --ink-2: #bcc5c1; --muted: #8b9591;
39     --line: #2e3432; --grid: #262b29;
40     --accent: #7fcdcd; --accent-ink: #0d1f1b;
41     --good-ink: #5fd35f; --warn-ink: #fab219; --bad-ink: #f07a7a;
42     --c1: #3987e5; --c2: #d95926; --c3: #199e70; --c4: #c98500;
43     --c5: #d55181; --c6: #008300; --c7: #9085e9; --c8: #e66767;
44     --ghost: #4d5552;
45   }
46 }
47 :root[data-theme="dark"] {
48   color-scheme: dark;
49   --bg: #101312; --surface: #1a1d1c; --surface-2: #232826;
50   --ink: #f1f4f2; --ink-2: #bcc5c1; --muted: #8b9591;
51   --line: #2e3432; --grid: #262b29;
52   --accent: #7fcdcd; --accent-ink: #0d1f1b;
53   --good-ink: #5fd35f; --warn-ink: #fab219; --bad-ink: #f07a7a;
54   --c1: #3987e5; --c2: #d95926; --c3: #199e70; --c4: #c98500;
55   --c5: #d55181; --c6: #008300; --c7: #9085e9; --c8: #e66767;
56   --ghost: #4d5552;
57 }
58
59 * { box-sizing: border-box; }
60 html, body { margin: 0; }
61 body {
62   background: var(--bg);
63   color: var(--ink);
64   font: 400 14px/1.5 var(--sans);
65   padding-inline: 16px;
66   padding-block: 20px 48px;
67 }
68 .wrap { max-width: 1280px; margin: 0 auto; display: grid; gap: 16px; }
69 .num { font-family: var(--mono); font-variant-numeric: tabular-nums; }
70
71 /* Header */
72 header.top { display: flex; flex-wrap: wrap; align-items: end; justify-content: space-between; gap: 16px;
  padding-block: 4px 8px; }
```

```

73 .brand h1 { font: 700 clamp(26px, 4vw, 36px)/1.05 var(--display); margin: 0; letter-spacing: -0.02em; text-align: center; }
74 .brand p { margin: 4px 0 0; color: var(--ink-2); }
75 .controls { display: flex; flex-wrap: wrap; gap: 8px; align-items: center; }
76 label.inline { display: flex; flex-direction: column; gap: 2px; font-size: 11px; text-transform: uppercase; }
77 select, input, button { font: inherit; color: inherit; }
78 select, input {
79   background: var(--surface); border: 1px solid var(--line); border-radius: 8px;
80   padding: 7px 10px; min-height: 36px; color: var(--ink);
81 }
82 select:focus-visible, input:focus-visible, button:focus-visible { outline: 2px solid var(--accent); outline-offset: 2px; }
83 .btn {
84   border: 1px solid var(--line); background: var(--surface); color: var(--ink);
85   border-radius: 8px; padding: 7px 14px; min-height: 36px; cursor: pointer; font-weight: 500;
86 }
87 .btn:hover { background: var(--surface-2); }
88 .btn.primary { background: var(--accent); color: var(--accent-ink); border-color: var(--accent); }
89 .btn.primary:hover { filter: brightness(1.08); }
90 .btn.danger-armed { border-color: var(--bad); color: var(--bad-ink); }
91
92 .notice {
93   display: flex; flex-wrap: wrap; gap: 8px 16px; align-items: center; justify-content: space-between;
94   background: var(--surface-2); border: 1px dashed var(--line); border-radius: 10px; padding: 10px 14px;
95   color: var(--ink-2);
96 }
97 .notice strong { color: var(--ink); }
98
99 /* KPI tiles */
100 .kpi { display: grid; grid-template-columns: repeat(4, minmax(0, 1fr)); gap: 16px; }
101 .kpi { background: var(--surface); border: 1px solid var(--line); border-radius: 12px; padding: 16px 18px; }
102 .kpi .label { font-size: 11px; text-transform: uppercase; letter-spacing: 0.07em; color: var(--muted); font-weight: 600; }
103 .kpi .value { font: 500 clamp(22px, 2.6vw, 30px)/1.1 var(--mono); font-variant-numeric: tabular-nums; letter-spacing: -0.02em; }
104 .kpi .sub { color: var(--ink-2); font-size: 13px; display: flex; gap: 6px; align-items: center; flex-wrap: wrap; }
105 .meter { height: 6px; border-radius: 3px; background: var(--surface-2); overflow: hidden; }
106 .meter > span { display: block; height: 100%; border-radius: 3px; background: var(--accent); }
107
108 .pill { display: inline-flex; align-items: center; gap: 5px; padding: 1px 8px 1px 6px; border-radius: 999px; font-size: 12px; font-weight: 600; border: 1px solid currentColor; line-height: 1.6; white-space: nowrap; }
109 .pill::before { content: ""; width: 7px; height: 7px; border-radius: 50%; background: var(--dot); }
110 .pill.good { color: var(--good-ink); --dot: var(--good); }
111 .pill.warn { color: var(--warn-ink); --dot: var(--warn); }
112 .pill.bad { color: var(--bad-ink); --dot: var(--bad); }
113 .delta.up { color: var(--bad-ink); } .delta.down { color: var(--good-ink); }
114
115 /* Panels */
116 .grid { display: grid; grid-template-columns: repeat(12, minmax(0, 1fr)); gap: 16px; }
117 .panel { background: var(--surface); border: 1px solid var(--line); border-radius: 12px; padding: 16px 18px; display: grid; gap: 12px; align-content: start; min-width: 0; }
118 .panel h2 { font: 600 16px/1.3 var(--display); margin: 0; letter-spacing: -0.01em; }
119 .panel .head { display: flex; justify-content: space-between; align-items: baseline; gap: 12px; flex-wrap: wrap; }
120 .panel .hint { color: var(--muted); font-size: 12px; }
121 .span-8 { grid-column: span 8; } .span-7 { grid-column: span 7; } .span-5 { grid-column: span 5; } .span-4 { grid-column: span 4; }
122 .stack { display: grid; gap: 16px; align-content: start; min-width: 0; }
123 .chart-box { position: relative; height: 280px; }
124 .chart-box.short { height: 200px; }
125 .chart-box.tall { height: 400px; }
126 .chart-box.mid { height: 340px; }
127
128 .legend { display: flex; flex-wrap: wrap; gap: 6px 14px; font-size: 12px; color: var(--ink-2); }
129 .legend span { display: inline-flex; align-items: center; gap: 6px; }
130 .swatch { width: 10px; height: 10px; border-radius: 3px; background: var(--c); flex: none; }
131 .swatch.line { height: 2px; border-radius: 1px; width: 14px; }
132 .swatch.dash { height: 0; width: 14px; border-top: 2px dashed var(--c); background: none; }
133
134 /* Donut */
135 .donut-wrap { position: relative; height: 220px; }
136 .donut-center { position: absolute; inset: 0; display: grid; place-content: center; text-align: center; pointer-events: none; }
137 .donut-center .v { font: 500 20px/1.1 var(--mono); font-variant-numeric: tabular-nums; }
138 .donut-center .l { font-size: 11px; color: var(--muted); text-transform: uppercase; letter-spacing: 0.07em; }

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```

    m; }
138 .cat-list { display: grid; gap: 2px; margin: 0; padding: 0; list-style: none; }
139 .cat-list li { display: grid; grid-template-columns: auto 1fr auto auto; gap: 10px; align-items: center;
    padding: 5px 4px; border-radius: 6px; }
140 .cat-list li:hover { background: var(--surface-2); }
141 .cat-list .amt { font-family: var(--mono); font-variant-numeric: tabular-nums; }
142 .cat-list .pct { font-family: var(--mono); font-variant-numeric: tabular-nums; color: var(--muted); width
    : 3.5em; text-align: right; }
143
144 /* Budgets */
145 .budgets { display: grid; gap: 12px; margin: 0; padding: 0; list-style: none; }
146 .budgets li { display: grid; gap: 5px; }
147 .budgets .row { display: flex; justify-content: space-between; align-items: center; gap: 8px; flex-wrap:
    wrap; }
148 .budgets .name { display: inline-flex; align-items: center; gap: 8px; font-weight: 500; }
149 .budgets .fig { font-family: var(--mono); font-variant-numeric: tabular-nums; font-size: 13px; color: var
    (--ink-2); display: inline-flex; align-items: center; gap: 4px; }
150 .budgets input { width: 92px; min-height: 28px; padding: 2px 6px; font-family: var(--mono); font-size: 13
    px; text-align: right; }
151 .bar { position: relative; height: 8px; border-radius: 4px; background: var(--surface-2); overflow: hidde
    n; }
152 .bar > span { position: absolute; inset: 0 auto 0 0; border-radius: 4px; background: var(--c); }
153 .bar.over > span { background: repeating-linear-gradient(135deg, var(--c) 0 6px, color-mix(in srgb, var(-
    -c) 60%, var(--surface)) 6px 9px); }
154
155 /* Form */
156 form.add { display: grid; grid-template-columns: 1fr 1fr; gap: 10px; }
157 form.add .full { grid-column: 1 / -1; }
158 form.add label { display: grid; gap: 4px; font-size: 12px; color: var(--ink-2); font-weight: 500; }
159 form.add input, form.add select { width: 100%; }
160 .form-msg { font-size: 13px; min-height: 1.5em; color: var(--ink-2); }
161 .form-msg.err { color: var(--bad-ink); }
162
163 /* Transactions */
164 .filters { display: flex; flex-wrap: wrap; gap: 6px; }
165 .chip { display: inline-flex; align-items: center; gap: 6px; border: 1px solid var(--line); background: v
    ar(--surface); color: var(--ink-2); border-radius: 999px; padding: 3px 10px; font-size: 12px; cursor: point
    er; min-height: 28px; }
166 .chip[aria-pressed="true"] { background: var(--ink); color: var(--bg); border-color: var(--ink); }
167 .chip .swatch { width: 8px; height: 8px; border-radius: 50%; }
168 .table-scroll { overflow-x: auto; max-height: 460px; overflow-y: auto; border-top: 1px solid var(--line);
    }
169 table { width: 100%; border-collapse: collapse; font-size: 13px; }
170 th, td { text-align: left; padding: 8px 6px; border-bottom: 1px solid var(--grid); white-space: nowrap; }
171 th { position: sticky; top: 0; background: var(--surface); font-size: 11px; text-transform: uppercase; le
    tter-spacing: 0.06em; color: var(--muted); font-weight: 600; }
172 td.amt, th.amt { text-align: right; font-family: var(--mono); font-variant-numeric: tabular-nums; }
173 td.note { white-space: normal; min-width: 140px; color: var(--ink-2); }
174 .cat-tag { display: inline-flex; align-items: center; gap: 6px; }
175 .del { border: 0; background: none; color: var(--muted); cursor: pointer; font-size: 16px; line-height: 1
    ; padding: 4px 8px; border-radius: 6px; }
176 .del:hover { color: var(--bad-ink); background: var(--surface-2); }
177 .empty { color: var(--muted); padding: 16px 4px; text-align: center; }
178 .foot { display: flex; justify-content: space-between; gap: 8px; flex-wrap: wrap; color: var(--ink-2); fo
    nt-size: 13px; }
179
180 @media (max-width: 1000px) {
181     .span-8, .span-7, .span-5, .span-4 { grid-column: 1 / -1; }
182     .kpis { grid-template-columns: repeat(2, minmax(0, 1fr)); }
183 }
184 @media (max-width: 460px) {
185     .kpis { grid-template-columns: 1fr; }
186     form.add { grid-template-columns: 1fr; }
187     .chart-box, .chart-box.tall, .chart-box.mid { height: 260px; }
188 }
189 @media (prefers-reduced-motion: reduce) { * { transition: none !important; animation: none !important; }
    }
190 </style>
191 <!-- ARTIFACT-HEAD-END -->
192 </head>

```

HTML: page structure

```

193 <body>
194 <!-- ARTIFACT-BODY-START -->
195 <div class="wrap">
196     <header class="top">
197         <div class="brand">
198             <h1>Pocket Ledger</h1>

```

```

199     <p>Where your money went, by month and by category.</p>
200 </div>
201 <div class="controls">
202     <label class="inline" for="monthSel">Month
203     <select id="monthSel"></select>
204 </label>
205     <label class="inline" for="curSel">Currency
206     <select id="curSel">
207         <option value="INR">₹ INR</option>
208         <option value="USD">$ USD</option>
209         <option value="EUR">€ EUR</option>
210         <option value="GBP">£ GBP</option>
211     </select>
212 </label>
213 </div>
214 </header>
215
216 <div class="notice" id="sampleNotice" hidden>
217     <span><strong>You're looking at sample data.</strong> Add your own expenses below, or clear the sample
218     to start fresh. Everything is saved in this browser only.</span>
219     <button class="btn" id="clearBtn" type="button">Clear sample data</button>
220 </div>
221 <section class="kpi" aria-label="Month summary">
222     <div class="kpi">
223         <span class="label">Spent this month</span>
224         <span class="value" id="kSpent"></span>
225         <span class="sub" id="kSpentSub"></span>
226     </div>
227     <div class="kpi">
228         <span class="label">Budget remaining</span>
229         <span class="value" id="kLeft"></span>
230         <div class="meter" aria-hidden="true"><span id="kMeter" style="width:0%"></span></div>
231         <span class="sub" id="kLeftSub"></span>
232     </div>
233     <div class="kpi">
234         <span class="label">Daily average</span>
235         <span class="value" id="kDaily"></span>
236         <span class="sub" id="kDailySub"></span>
237     </div>
238     <div class="kpi">
239         <span class="label">Top category</span>
240         <span class="value" id="kTop" style="font-family:var(--display);font-weight:600;letter-spacing:-0.01e
241         m"></span>
242         <span class="sub" id="kTopSub"></span>
243     </div>
244 </section>
245 <section class="grid">
246     <div class="panel span-8">
247         <div class="head"><h2>Six-month spending by category</h2><span class="hint">Hover a bar for the break
248         down</span></div>
249         <div class="legend" id="catLegend"></div>
250         <div class="chart-box tall"><canvas id="trendChart" aria-label="Stacked bar chart of monthly spending
251         by category"></canvas></div>
252     </div>
253     <div class="panel span-4">
254         <div class="head"><h2>Where it went</h2><span class="hint" id="donutHint"></span></div>
255         <div class="donut-wrap">
256             <canvas id="donutChart" aria-label="Donut chart of spending share by category"></canvas>
257             <div class="donut-center"><span class="v num" id="donutTotal"></span><span class="l">total</span></div>
258         </div>
259     </div>
260     <div class="cat-list" id="catList"></div>
261 </div>
262 <div class="panel span-7">
263     <div class="head"><h2>Running total vs. budget pace</h2><span class="hint">Cumulative spend through t
264     he month</span></div>
265     <div class="legend">
266         <span><i class="swatch line" style="--c:var(--c1)"></i>This month</span>
267         <span><i class="swatch line" style="--c:var(--ghost)"></i>Previous month</span>
268         <span><i class="swatch dash" style="--c:var(--muted)"></i>Budget pace</span>
269     </div>
270     <div class="chart-box mid"><canvas id="paceChart" aria-label="Line chart of cumulative spending again
271     st budget pace"></canvas></div>
272 </div>
273 <div class="panel span-5">
274     <div class="head"><h2>Category budgets</h2><span class="hint">Edit a limit to change it</span></div>

```

```

271     <ul class="budgets" id="budgetList"></ul>
272 </div>
273
274 <div class="stack span-4">
275   <div class="panel">
276     <h2>Add an expense</h2>
277     <form class="add" id="addForm" novalidate>
278       <label for="fAmount">Amount
279       <input id="fAmount" type="number" inputmode="decimal" min="0" step="0.01" placeholder="0.00" re
280       </label>
281       <label for="fDate">Date
282       <input id="fDate" type="date" required>
283       </label>
284       <label class="full" for="fCat">Category
285       <select id="fCat"></select>
286       </label>
287       <label class="full" for="fNote">Note
288       <input id="fNote" type="text" maxlength="80" placeholder="e.g. Weekly groceries">
289       </label>
290       <button class="btn primary full" type="submit">Add expense</button>
291       <div class="form-msg full" id="formMsg" role="status"></div>
292     </form>
293   </div>
294   <div class="panel">
295     <div class="head"><h2>By day of week</h2><span class="hint">Selected month</span></div>
296     <div class="chart-box short"><canvas id="dowChart" aria-label="Bar chart of spending by day of week
297     </div>
298   </div>
299
300   <div class="panel span-8">
301     <div class="head">
302       <h2>Transactions</h2>
303       <input id="search" type="search" placeholder="Search notes" aria-label="Search notes" style="min-he
304       </div>
305       <div class="filters" id="filters" role="group" aria-label="Filter by category"></div>
306       <div class="table-scroll">
307         <table>
308           <thead><tr><th>Date</th><th>Category</th><th>Note</th><th class="amt">Amount</th><th><span class=
309           <tbody id="txBody"></tbody>
310         </table>
311       </div>
312       <div class="foot"><span id="txCount"></span><span id="txSum" class="num"></span></div>
313     </div>
314   </section>
315 </div>
316

```

JavaScript: data, charts and interactions

```

317 <script>
318 (() => {
319   const CATS = [
320     { id: "housing", name: "Housing", v: "--c1", budget: 18000 },
321     { id: "food", name: "Food & Dining", v: "--c2", budget: 12000 },
322     { id: "transport", name: "Transport", v: "--c3", budget: 6000 },
323     { id: "utilities", name: "Utilities", v: "--c4", budget: 4500 },
324     { id: "shopping", name: "Shopping", v: "--c5", budget: 6000 },
325     { id: "health", name: "Health", v: "--c6", budget: 3000 },
326     { id: "fun", name: "Entertainment", v: "--c7", budget: 3500 },
327     { id: "other", name: "Other", v: "--c8", budget: 2500 },
328   ];
329   const CAT = Object.fromEntries(CATS.map(c => [c.id, c]));
330   const KEY = "pocket-ledger-v1";
331   const LOCALES = { INR: "en-IN", USD: "en-US", EUR: "en-IE", GBP: "en-GB" };
332
333   const pad = n => String(n).padStart(2, "0");
334   const ymd = d => `${d.getFullYear()}-${pad(d.getMonth() + 1)}-${pad(d.getDate())}`;
335   const ym = d => `${d.getFullYear()}-${pad(d.getMonth() + 1)}`;
336   const today = new Date();
337   const uid = () => Math.random().toString(36).slice(2, 10);
338
339   // ----- Sample data (deterministic) -----
340   function mulberry32(a) { return () => { a |= 0; a = a + 0x6D2B79F5 | 0; let t = Math.imul(a ^ a >>> 15, 1
341   | a); t = t + Math.imul(t ^ t >>> 7, 61 | t) ^ t; return ((t ^ t >>> 14) >>> 0) / 4294967296; }; }
342   function buildSample() {

```

```

342     const r = mulberry32(42);
343     const rand = (lo, hi) => Math.round((lo + r() * (hi - lo)) / 10) * 10;
344     const out = [];
345     for (let m = 5; m >= 0; m--) {
346         const first = new Date(today.getFullYear(), today.getMonth() - m, 1);
347         const dim = new Date(first.getFullYear(), first.getMonth() + 1, 0).getDate();
348         const lastDay = m === 0 ? today.getDate() : dim;
349         const add = (day, cat, amount, note) => {
350             if (day > lastDay) return;
351             out.push({ id: uid(), date: ymd(new Date(first.getFullYear(), first.getMonth(), day)), cat, amount,
352 ↪ note });
353         };
354         const any = () => 1 + Math.floor(r() * dim);
355         add(1, "housing", 18000, "Rent");
356         add(5, "utilities", rand(1600, 2700), "Electricity bill");
357         add(10, "utilities", 799, "Home internet");
358         add(12, "utilities", 399, "Mobile recharge");
359         for (let i = 0; i < 5; i++) add(any(), "food", rand(700, 2300), ["Groceries", "Vegetables & fruit", "
360 ↪ Supermarket run", "Milk & bakery"][i % 4]);
361         for (let i = 0, n = 5 + Math.floor(r() * 5); i < n; i++) add(any(), "food", rand(220, 1300), ["Dinner
362 ↪ out", "Lunch with team", "Food delivery", "Coffee & snacks", "Weekend brunch"][i % 5]);
363         for (let i = 0, n = 8 + Math.floor(r() * 6); i < n; i++) add(any(), "transport", rand(80, 420), ["Cab
364 ↪ ride", "Auto", "Metro card top-up", "Parking"][i % 4]);
365         add(any(), "transport", rand(1500, 2400), "Fuel");
366         if (r() > 0.4) add(any(), "transport", rand(1500, 2400), "Fuel");
367         for (let i = 0, n = 2 + Math.floor(r() * 4); i < n; i++) add(any(), "shopping", rand(450, 3400), ["Cl
368 ↪ othes", "Household items", "Online order", "Shoes", "Gift"][i % 5]);
369         for (let i = 0, n = 1 + Math.floor(r() * 2); i < n; i++) add(any(), "health", rand(300, 2400), ["Phar
370 ↪ macy", "Doctor visit", "Gym membership"][i % 3]);
371         add(3, "fun", 649, "Streaming subscription");
372         for (let i = 0, n = 1 + Math.floor(r() * 4); i < n; i++) add(any(), "fun", rand(250, 1600), ["Movie t
373 ↪ ickets", "Concert", "Games", "Books"][i % 4]);
374         for (let i = 0, n = 1 + Math.floor(r() * 3); i < n; i++) add(any(), "other", rand(200, 1400), ["Hairc
375 ↪ ut", "Donation", "Stationery", "Laundry"][i % 4]);
376     }
377     return out;
378 }
379
380 // ----- State -----
381 function load() {
382     try { const s = JSON.parse(localStorage.getItem(KEY)); if (s && Array.isArray(s.expenses)) return s; }
383 ↪ catch (e) {}
384     return { expenses: buildSample(), budgets: Object.fromEntries(CATS.map(c => [c.id, c.budget])), currenc
385 ↪ y: "INR", sample: true };
386 }
387 function save() { try { localStorage.setItem(KEY, JSON.stringify(state)); } catch (e) {} }
388 const state = load();
389 CATS.forEach(c => { if (typeof state.budgets[c.id] !== "number") state.budgets[c.id] = c.budget; });
390 let month = ym(today);
391 let filterCat = "all";
392 let query = "";
393
394 let fmt, fmtShort;
395 function setCurrency() {
396     const loc = LOCALES[state.currency] || "en-US";
397     fmt = new Intl.NumberFormat(loc, { style: "currency", currency: state.currency, maximumFractionDigits:
398 ↪ 0 });
399     fmtShort = new Intl.NumberFormat("en-US", { style: "currency", currency: state.currency, notation: "com
400 ↪ pact", maximumFractionDigits: 1 });
401 }
402 const money = v => fmt.format(v);
403
404 // ----- Helpers -----
405 const $ = id => document.getElementById(id);
406 const css = v => getComputedStyle(document.documentElement).getPropertyValue(v).trim();
407 const color = c => css(CAT[c].v);
408 const inMonth = key => state.expenses.filter(e => e.date.startsWith(key));
409 const sum = arr => arr.reduce((a, e) => a + e.amount, 0);
410 const monthDate = key => { const [y, m] = key.split("-").map(Number); return new Date(y, m - 1, 1); };
411 const shiftMonth = (key, n) => { const d = monthDate(key); return ym(new Date(d.getFullYear(), d.getMonth
412 ↪ () + n, 1)); };
413 const monthLabel = (key, style = "long") => monthDate(key).toLocaleDateString("en-US", { month: style, ye
414 ↪ ar: "numeric" });
415 const daysIn = key => { const d = monthDate(key); return new Date(d.getFullYear(), d.getMonth() + 1, 0).g
416 ↪ etDate(); };
417 const byCat = list => { const o = Object.fromEntries(CATS.map(c => [c.id, 0])); list.forEach(e => { o[e.c
418 ↪ at] = (o[e.cat] || 0) + e.amount; }); return o; };
419 const esc = s => String(s).replace(/<[>"]'/g, ch => ({ "&": "&amp;", "<": "&lt;", ">": "&gt;", "'": "&qu
420 ↪ ot;", '"': "&#39;" }[ch]));

```

```

404   const totalBudget = () => CATS.reduce((a, c) => a + (state.budgets[c.id] || 0), 0);
405   const status = pct => pct > 1 ? ["bad", "Over budget"] : pct >= 0.8 ? ["warn", "Near limit"] : ["good", "
  ↳ On track"];
406
407   // ----- Controls -----
408   function fillMonths() {
409     const keys = new Set(state.expenses.map(e => e.date.slice(0, 7)));
410     keys.add(ym(today));
411     const sorted = [...keys].sort().reverse();
412     if (!sorted.includes(month)) month = sorted[0];
413     $("monthSel").innerHTML = sorted.map(k => `<option value="${k}"${k === month ? " selected" : ""}>${mont
  ↳ hLabel(k)}</option>`).join("");
414   }
415   $("monthSel").addEventListener("change", e => { month = e.target.value; render(); });
416   $("curSel").value = state.currency;
417   $("curSel").addEventListener("change", e => { state.currency = e.target.value; save(); setCurrency(); ren
  ↳ der(); });
418   $("fCat").innerHTML = CATS.map(c => `<option value="${c.id}">${c.name}</option>`).join("");
419   $("fDate").value = ymd(today);
420   $("catLegend").innerHTML = CATS.map(c => `<span><i class="swatch" style="--c:var(${c.v})"></i>${c.name}</
  ↳ span>`).join("");
421
422   $("addForm").addEventListener("submit", e => {
423     e.preventDefault();
424     const amount = parseFloat($("fAmount").value);
425     const date = $("fDate").value;
426     const msg = $("formMsg");
427     if (!(amount > 0)) { msg.className = "form-msg full err"; msg.textContent = "Enter an amount greater th
  ↳ an zero."; $("fAmount").focus(); return; }
428     if (!/^d{4}-d{2}-d{2}$/.test(date)) { msg.className = "form-msg full err"; msg.textContent = "Pick a
  ↳ date for this expense."; $("fDate").focus(); return; }
429     const cat = $("fCat").value;
430     const note = $("fNote").value.trim() || CAT[cat].name;
431     state.expenses.push({ id: uid(), date, cat, amount: Math.round(amount * 100) / 100, note });
432     save();
433     month = date.slice(0, 7);
434     msg.className = "form-msg full";
435     msg.textContent = `Added ${money(amount)} to ${CAT[cat].name}.`;
436     $("fAmount").value = ""; $("fNote").value = "";
437     render();
438   });
439
440   let clearArmed = false;
441   $("clearBtn").addEventListener("click", () => {
442     const b = $("clearBtn");
443     if (!clearArmed) { clearArmed = true; b.textContent = "Click again to clear"; b.classList.add("danger-a
  ↳ rmed"); setTimeout(() => { clearArmed = false; b.textContent = "Clear sample data"; b.classList.remove("dan
  ↳ ger-armed"); }, 4000); return; }
444     state.expenses = []; state.sample = false; save(); month = ym(today); render();
445   });
446
447   $("search").addEventListener("input", e => { query = e.target.value.toLowerCase(); renderTable(); });
448
449   // ----- Charts -----
450   const charts = {};
451   function baseOpts() {
452     const ink2 = css("--ink-2"), grid = css("--grid"), surface = css("--surface"), ink = css("--ink"), line
  ↳ = css("--line");
453     return {
454       responsive: true, maintainAspectRatio: false, animation: { duration: 400 },
455       plugins: {
456         legend: { display: false },
457         tooltip: {
458           backgroundColor: surface, titleColor: ink, bodyColor: ink2, borderColor: line, borderWidth: 1,
459           padding: 10, cornerRadius: 8, boxPadding: 4, usePointStyle: true,
460           titleFont: { family: "IBM Plex Sans", weight: "600" }, bodyFont: { family: "IBM Plex Mono" },
461         },
462       },
463       scales: {
464         x: { grid: { display: false }, border: { color: line }, ticks: { color: ink2, font: { family: "IBM
  ↳ Plex Sans", size: 12 } } },
465         y: { beginAtZero: true, grid: { color: grid }, border: { display: false }, ticks: { color: ink2, fo
  ↳ nt: { family: "IBM Plex Mono", size: 11 }, callback: v => fmtShort.format(v), maxTicksLimit: 6 } },
466       },
467     };
468   }
469   function makeChart(id, cfg) { if (charts[id]) charts[id].destroy(); charts[id] = new Chart$(id, cfg); }
470
471   function renderTrend() {

```

```

472 const keys = Array.from({ length: 6 }, (_, i) => shiftMonth(month, i - 5));
473 const surface = css("--surface");
474 const datasets = CATS.map((c, i) => ({
475   label: c.name,
476   data: keys.map(k => byCat(inMonth(k))[c.id]),
477   backgroundColor: color(c.id),
478   borderColor: surface,
479   borderWidth: { top: 2, left: 0, right: 0, bottom: 0 },
480   borderRadius: i === CATS.length - 1 ? { topLeft: 4, topRight: 4 } : 0,
481   borderSkipped: "bottom",
482   maxBarThickness: 44,
483   stack: "s",
484 }));
485 const opts = baseOpts();
486 opts.scales.x.stacked = true; opts.scales.y.stacked = true;
487 opts.interaction = { mode: "index", intersect: false };
488 opts.plugins.tooltip.itemSort = (a, b) => b.raw - a.raw;
489 opts.plugins.tooltip.filter = item => item.raw > 0;
490 opts.plugins.tooltip.callbacks = {
491   label: ctx => ` ${ctx.dataset.label}: ${money(ctx.raw)} `,
492   footer: items => `Total: ${money(items.reduce((a, i) => a + i.raw, 0))}`,
493 };
494 opts.plugins.tooltip.footerFont = { family: "IBM Plex Mono", weight: "600" };
495 opts.plugins.tooltip.footerColor = css("--ink");
496 makeChart("trendChart", { type: "bar", data: { labels: keys.map(k => monthDate(k).toLocaleDateString("en-US", { month: "short" } ) + (k === month ? " ●" : "")), datasets }, options: opts });
497 }
498
499 function renderDonut(cats, total) {
500   const surface = css("--surface");
501   const order = CATS.filter(c => cats[c.id] > 0);
502   makeChart("donutChart", {
503     type: "doughnut",
504     data: { labels: order.map(c => c.name), datasets: [{ data: order.map(c => cats[c.id]), backgroundColor: order.map(c => color(c.id)), borderColor: surface, borderWidth: 2, hoverOffset: 6, borderRadius: 3 } ] },
505     options: {
506       responsive: true, maintainAspectRatio: false, cutout: "70%", layout: { padding: 6 },
507       plugins: {
508         legend: { display: false },
509         tooltip: { ...baseOpts().plugins.tooltip, callbacks: { label: ctx => ` ${money(ctx.raw)} · ${((ctx.x.raw / total) * 100).toFixed(1)}%` } },
510       },
511     },
512   });
513   $("donutTotal").textContent = money(total);
514   $("donutHint").textContent = monthLabel(month, "short");
515   const sorted = [...CATS].sort((a, b) => cats[b.id] - cats[a.id]);
516   $("catList").innerHTML = sorted.map(c => `<li><i class="swatch" style="--c:var(${c.v})"></i><span>${c.name}</span><span class="amt">${money(cats[c.id])}</span><span class="pct">${total ? Math.round(cats[c.id] / total * 100) : 0}%</span></li>`).join("");
517 }
518
519 function renderPace() {
520   const dim = daysIn(month);
521   const isCurrent = month === ym(today);
522   const lastDay = isCurrent ? today.getDate() : dim;
523   const cum = key => {
524     const n = daysIn(key); const daily = Array(n).fill(0);
525     inMonth(key).forEach(e => { daily[Number(e.date.slice(8, 10)) - 1] += e.amount; });
526     let run = 0; return daily.map(v => (run += v));
527   };
528   const cur = cum(month).map((v, i) => (i < lastDay ? v : null));
529   const prevRaw = cum(shiftMonth(month, -1));
530   const prev = Array.from({ length: dim }, (_, i) => prevRaw[Math.min(i, prevRaw.length - 1)]);
531   const tb = totalBudget();
532   const pace = Array.from({ length: dim }, (_, i) => Math.round(tb * (i + 1) / dim));
533   const c1 = css("--c1"), ghost = css("--ghost"), muted = css("--muted"), surface = css("--surface");
534   const opts = baseOpts();
535   opts.interaction = { mode: "index", intersect: false };
536   opts.scales.x.ticks.maxTicksLimit = 10;
537   opts.scales.x.ticks.autoSkip = true;
538   opts.plugins.tooltip.callbacks = {
539     title: items => monthDate(month).toLocaleDateString("en-US", { month: "short" } ) + " " + items[0].label,
540     label: ctx => ctx.raw == null ? null : ` ${ctx.dataset.label}: ${money(ctx.raw)} `,
541   };
542   opts.plugins.tooltip.filter = item => item.raw != null;
543   makeChart("paceChart", {
544     type: "line",

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545     data: {
546       labels: Array.from({ length: dim }, (_, i) => i + 1),
547       datasets: [
548         { label: "This month", data: cur, borderColor: c1, backgroundColor: c1 + "22", fill: "origin", bo
549         rderWidth: 2, tension: 0.25, pointRadius: cur.map((v, i) => i === lastDay - 1 ? 5 : 0), pointBackgroundColo
550         r: c1, pointBorderColor: surface, pointBorderWidth: 2, pointHoverRadius: 5, order: 1 },
551         { label: "Previous month", data: prev, borderColor: ghost, borderWidth: 2, tension: 0.25, pointRa
552         dius: 0, pointHoverRadius: 4, fill: false, order: 2 },
553         { label: "Budget pace", data: pace, borderColor: muted, borderDash: [5, 4], borderWidth: 1.5, poi
554         ntRadius: 0, pointHoverRadius: 0, fill: false, order: 3 },
555       ],
556     },
557     options: opts,
558   });
559 }
560
561 function renderDow(list) {
562   const names = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"];
563   const vals = Array(7).fill(0);
564   list.forEach(e => { const [y, m, d] = e.date.split("-").map(Number); vals[(new Date(y, m - 1, d).getDay
565   () + 6) % 7] += e.amount; });
566   const opts = baseOpts();
567   opts.scales.y.ticks.maxTicksLimit = 4;
568   opts.plugins.tooltip.callbacks = { label: ctx => ` ${money(ctx.raw)} ` };
569   makeChart("dowChart", { type: "bar", data: { labels: names, datasets: [{ label: "Spent", data: vals, ba
570   ckgroundColor: css("--c1"), borderRadius: { topLeft: 4, topRight: 4 }, borderSkipped: "bottom", maxBarThick
571   ness: 28 } ] }, options: opts });
572 }
573
574 // ----- KPI / budgets / table -----
575 function renderKpis(list, cats, total) {
576   const prevTotal = sum(inMonth(shiftMonth(month, -1)));
577   $("kSpent").textContent = money(total);
578   if (prevTotal > 0) {
579     const d = (total - prevTotal) / prevTotal;
580     const isCurrent = month === ym(today);
581     $("kSpentSub").innerHTML = `<span class="delta ${d > 0 ? "up" : "down"} num">${d > 0 ? "▲" : "▼"} ${M
582     ath.abs(d * 100).toFixed(0)}%</span> vs ${money(prevTotal)} ${isCurrent ? "all of" : "in"} ${monthDate(shif
583     tMonth(month, -1)).toLocaleDateString("en-US", { month: "short" })}`;
584   } else $("kSpentSub").textContent = `${list.length} transactions`;
585
586   const tb = totalBudget(); const pct = tb ? total / tb : 0; const [cls, lbl] = status(pct);
587   $("kLeft").textContent = money(tb - total);
588   $("kMeter").style.width = Math.min(100, pct * 100).toFixed(1) + "%";
589   $("kMeter").style.background = `var(--${cls})`;
590   $("kLeftSub").innerHTML = `<span class="pill ${cls}">${lbl}</span><span class="num">${Math.round(pct *
591   100)}% of ${money(tb)}</span>`;
592
593   const isCurrent = month === ym(today);
594   const days = isCurrent ? today.getDate() : daysIn(month);
595   $("kDaily").textContent = money(days ? total / days : 0);
596   if (isCurrent) {
597     const proj = days ? total / days * daysIn(month) : 0;
598     $("kDailySub").innerHTML = `Projected month: <span class="num">${money(proj)}</span>`;
599   } else $("kDailySub").textContent = `Across ${days} days`;
600
601   const top = [...CATS].sort((a, b) => cats[b.id] - cats[a.id])[0];
602   if (total > 0) {
603     $("kTop").textContent = top.name;
604     $("kTopSub").innerHTML = `<i class="swatch" style="--c:var(${top.v})"></i><span class="num">${money(c
605     ats[top.id])}</span> · ${Math.round(cats[top.id] / total * 100)}% of spending`;
606   } else { $("kTop").textContent = "-"; $("kTopSub").textContent = "No spending yet"; }
607 }
608
609 function renderBudgets(cats) {
610   $("budgetList").innerHTML = CATS.map(c => {
611     const b = state.budgets[c.id] || 0, s = cats[c.id]; const pct = b ? s / b : (s ? 2 : 0); const [cls,
612     lbl] = status(pct);
613     return `<li>
614       <div class="row">
615         <span class="name"><i class="swatch" style="--c:var(${c.v})"></i>${c.name} <span class="pill ${cl
616         s}">${lbl}</span></span>
617         <span class="fig">${money(s)} /
618         <input type="number" min="0" step="100" id="bud-${c.id}" data-cat="${c.id}" value="${b}" aria-l
619         abel="${c.name} monthly budget"></span>
620       </div>
621       <div class="bar${pct > 1 ? " over" : ""}" style="--c:var(${c.v})" role="img" aria-label="${c.name}:
622       ${Math.round(pct * 100)}% of budget used"><span style="width:${Math.min(100, pct * 100).toFixed(1)}%></sp
623       an></div>

```

```

608     </li>`;
609   }).join("");
610   $("budgetList").querySelectorAll("input").forEach(inp => inp.addEventListener("change", e => {
611     const v = Math.max(0, parseFloat(e.target.value) || 0);
612     state.budgets[e.target.dataset.cat] = v; save(); render();
613   }));
614 }
615
616 function renderFilters() {
617   const chips = [{ id: "all", name: "All" }, ...CATS];
618   $("filters").innerHTML = chips.map(c => `<button type="button" class="chip" data-cat="${c.id}" aria-pre
619     ssed="${filterCat === c.id}">${c.v ? `<i class="swatch" style="--c:var(${c.v})"></i>` : ""}${c.name}</butto
620     n>`).join("");
621   $("filters").querySelectorAll(".chip").forEach(b => b.addEventListener("click", () => { filterCat = b.d
622     ataset.cat; renderFilters(); renderTable(); }));
623 }
624
625 function renderTable() {
626   let list = inMonth(month).filter(e => (filterCat === "all" || e.cat === filterCat) && (!query || e.note
627     .toLowerCase().includes(query)));
628   list.sort((a, b) => b.date.localeCompare(a.date) || b.amount - a.amount);
629   $("tbody").innerHTML = list.length ? list.map(e => {
630     const [y, m, d] = e.date.split("-").map(Number);
631     const ds = new Date(y, m - 1, d).toLocaleDateString("en-US", { weekday: "short", month: "short", day:
632     "numeric" });
633     return `<tr><td>${ds}</td><td><span class="cat-tag"><i class="swatch" style="--c:var(${CAT[e.cat].v})
634     "></i>${CAT[e.cat].name}</span></td><td class="note">${esc(e.note)}</td><td class="amt">${money(e.amount)}<
635     /td><td><button class="del" data-id="${e.id}" aria-label="Delete ${esc(e.note)}" title="Delete">x</button><
636     /td></tr>`;
637   }).join("") : `<tr><td colspan="5" class="empty">No expenses match. Add one with the form, or change th
638     e filter.</td></tr>`;
639   $("tbody").querySelectorAll(".del").forEach(b => b.addEventListener("click", () => {
640     state.expenses = state.expenses.filter(e => e.id !== b.dataset.id); save(); render();
641   }));
642   $("txCount").textContent = `${list.length} transaction${list.length === 1 ? "" : "s"}`;
643   $("txSum").textContent = `Total ${money(sum(list))}`;
644 }
645
646 function render() {
647   fillMonths();
648   $("sampleNotice").hidden = !state.sample;
649   const list = inMonth(month), cats = byCat(list), total = sum(list);
650   renderKpis(list, cats, total);
651   renderTrend();
652   renderDonut(cats, total);
653   renderPace();
654   renderBudgets(cats);
655   renderDow(list);
656   renderFilters();
657   renderTable();
658 }
659
660 function start() {
661   setCurrency();
662   if (window.Chart) { Chart.defaults.font.family = "IBM Plex Sans, system-ui, sans-serif"; }
663   render();
664 }
665
666 // Re-render charts when the theme changes so they pick up the new tokens.
667 const rerender = () => { if (window.Chart) render(); };
668 try { matchMedia("(prefers-color-scheme: dark)").addEventListener("change", rerender); } catch (e) {}
669 new MutationObserver(rerender).observe(document.documentElement, { attributes: true, attributeFilter: ["d
670   ata-theme" ] });
671
672 if (document.fonts && document.fonts.ready) document.fonts.ready.then(() => { if (charts.trendChart) rend
673   er(); });
674 start();
675 })();
676 </script>
677 <!-- ARTIFACT-BODY-END -->
678 </body>
679 </html>

```